

Alaiy Hiring Challenge

ERPNext + Shopify Integration

Background

We operate an ecommerce business using Shopify and are evaluating ERPNext as a potential operational backbone for inventory, orders, purchasing, and business workflows.

ERPNext is built on the Frappe Framework, an open-source Python-based application framework. The ERPNext ecosystem contains a large number of community apps, integrations, and extensions. However, compatibility varies significantly across ERPNext versions, and not all integrations are maintained for newer releases.

This assignment is designed to evaluate your ability to:

- Learn unfamiliar technologies quickly
- Use AI tools effectively during development and research
- Research and validate technical decisions
- Design and implement integrations
- Solve practical problems independently
- Produce clear documentation

Objective

Determine the most suitable ERPNext version for an ecommerce business that relies heavily on integrations and ecosystem support.

Deploy ERPNext, connect it to an existing Shopify store, and demonstrate successful synchronization of ecommerce data.

The focus is not only on achieving a working integration but also on demonstrating how you approach unknown systems and make technical decisions.

Task 1: ERPNext Ecosystem Research

Research the ERPNext and Frappe ecosystem.

Produce a recommendation identifying:

- Which ERPNext version should be used
- Why that version was selected
- What alternatives were considered
- What integrations and apps are available
- Compatibility considerations

- Upgrade and maintenance implications

Your recommendation should prioritize:

- Ecosystem support
- Availability of integrations
- Stability
- Community adoption
- Long-term maintainability

Do not assume that the latest version is the correct choice.

Task 2: ERPNext Deployment

Deploy a working ERPNext instance on the provided infrastructure.

Document:

- Deployment approach
- Installation steps
- Configuration decisions
- Problems encountered
- Solutions implemented

The final system should be accessible and usable.

Task 3: Shopify Integration

Connect ERPNext with the provided Shopify store.

Synchronize as many of the following as possible:

Products

- Products
- Variants
- Images
- Pricing
- Inventory information

Orders

- Orders
- Customers
- Order status

- Fulfillment status
- Cancellations

Inventory

- Inventory levels
- Stock updates
- Inventory reconciliation approach

Document all mappings between Shopify entities and ERPNext entities.

Task 4: Data Validation

Where suitable data does not already exist in Shopify, create test records and demonstrate synchronization.

Examples:

- Multi-variant products
- Inventory changes
- Orders
- Cancellations
- Refunds
- Discounted orders
- Out-of-stock scenarios

Demonstrate that the data appears correctly inside ERPNext.

Task 5: Integration Assessment

Evaluate the quality of the Shopify ↔ ERPNext integration.

Answer questions such as:

- What works well?
- What limitations exist?
- What would need customization?
- What would break at scale?
- What monitoring would be required?
- How would you handle synchronization failures?

Bonus Challenges

Additional ecommerce integrations are encouraged.

Potential examples:

- Supplier inventory synchronization
- Shipping providers
- Payment gateways
- Marketplace integrations
- Automated stock reconciliation
- ERPNext custom apps
- Workflow automation
- Reporting dashboards

For each bonus integration:

- Explain business value
- Explain implementation approach
- Estimate complexity

Working implementations receive higher credit than theoretical proposals.

Evaluation Criteria

We are not evaluating prior ERPNext experience.

We are evaluating:

- Ability to learn unfamiliar systems quickly
- Quality of technical research
- Problem-solving ability
- Integration and API skills
- Use of AI-assisted development workflows
- Quality of implementation
- Documentation quality
- Engineering judgment

A candidate who can independently understand a new ecosystem, make reasonable decisions, and deliver a working solution will score highly even if every feature is not completed.

Optional Discussion

Include a short section describing:

"If you were responsible for the ecommerce technology stack for the next 12 months, how would you structure ERPNext, Shopify, integrations, inventory synchronization, and operational workflows?"

There is no single correct answer. We are interested in your reasoning.